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A structural equation modeling framework for studying sustainable freight transportation indicators in emerging economies

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ABSTRACT

The movement of commodities in emerging economies is largely facilitated by freight transit. The major obstacle to developing sustainable freight transportation systems in emerging economies are multiple dynamic uncertainties and inherent sustainability risks. This study determines the major indicators responsible for the sustainability of freight transportation. Initially, 16 Sustainable Transportation Indicators (STIs) are identified through literature and discussed with experts under five sustainability parameters: reputation, overall service quality, resilience, economic model, and job satisfaction. The identified STIs are categorized under the economic, social, environmental, and risk categories. Later, a hypothetical conceptual Structural Equation Model (SEM) is employed to analyze and verify the data. The results from the study show that STIs such as Reputation, Employment Benefits, and Infrastructure and Technology are crucial for the development of sustainable freight transport in emerging economies as they can enhance the efficiency and productivity of transportation systems and, hence their importance must be considered for future deployment of freight transportation in these economies.

1. Introduction

In the transportation sector, the flow of goods connects suppliers, transporters, and various clients [1]. The impacts of freight transportation on the economy, society, and environment are considerable and long-lasting [2]. The majority of urban and rural activities rely on transport and distribution, which is a significant employer and an essential factor in economic growth of the nation [3,4]. The functionality of logistics and supply chain system depends on freight transportation. The intent of the transportation system is to efficiently move the optimum amount and quality of commodities, finished goods, raw materials, and more from the source to the destination using a variety of modes, including road, rail, air, and water [5]. Freight transportation makes up between one-third to nearly two-thirds of the logistics budget, or around 10% to 20% of the cost of a commodity [6].

Consequently, the development of a nation depends largely on freight movements [7]. Also, it has long-term consequences on sustainability, including elevated noise pollution, air pollution, greenhouse emissions, traffic congestion, and accidents [8]. The freightage activities are responsible for around 80% to 90% of the carbon emissions in logistics operations [9]. This brings in the requirement for the freight transportation to be sustainable [10]. Consequently, the development of sustainable transportation has become a priority for policymakers in both developed and emerging economies [11]. The Brundtland Commission defines sustainable development as progress that satisfies current demands without endangering the capacity of future generations to meet these needs. Mostly, Freight Transportation Sustainability comprises of environmental, social, and economical aspects [12]. A sustainable transportation system should fulfills the requirements of the individuals without posing a threat to society. Also, it should be efficient, effective and affordable. Moreover, it should minimize waste and emissions and contributes towards carbon neutrality of the planet [13].

The establishment of a sustainable transport system is complicated due to various dynamic constraints and underlying sustainability risks. Sustainability risks are inherent risks that have direct consequences on the sustainable aspects of freight transportation system [14] and can affect the adoption of the triple bottom line framework [15]. Sustainability risks arise as a result of uncertainties in delivery procedures, demand, infrastructure, cost, and many other factors. The transportation authorities are primarily concerned with cost minimization and profit maximization that poses a risk to Sustainable Freight Transportation (SFT). Nonetheless, as environmental awareness and Government regulations increase, businesses are being forced to work

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in more viable ways for transportation systems to evolve [16]. SFT can integrate environmental protection with economic benefits such as saving, higher sales, retaining customers, quality enhancement, and greater prosperity [17,18].

In this study, for the identified Sustainable Transportation Indicators (STIs) (under categories: economic, social, environmental, and risk), a hypothetical Structural Equation Model (SEM) is developed. This SEM aims to examine every STI affecting sustainability and the numerous aspects connected to it including reputation, overall service quality, resilience, the economic model, and job satisfaction. SEM was applied for conceptual model validation and their influence on the sustainability indicators based on SFT in emerging economy.

The study is distributed as follows: Section 2 reflects the literature review on STIs, and Section 3 proposes a SEM for conceptual model validation and for identifying the key aspects of the sustainability of the nation's freight transportation system. In Section 4, the proposed model is tested using data obtained from the survey. Section 5 provides discussion and managerial implications and finally, Section 6 provides the conclusion of the study.

2. Literature review

In past few years, sustainability has matured into a vital element of modern business operations [19,20]. The leading cause for transportation corporations opting for sustainability is the enhancement of strict government regulations, increasing environmental concern, global warming, ecological issues, social issues, and importantly the focus of companies on maintaining a good reputation [15,21,22]. As a result, sustainability has become vital to businesses and organizations. Integrating sustainability aspect in the freight transportation industry is important to prevent adverse impacts on people and the environment [5]. Indicators are specific variables that can be measured, evaluated, and easily understandable [23,24]. Further, they must present accurate information in a simple and balanced manner to be understandable [25,26]. In this study, STIs are identified to investigate the sustainability of the nation's freight transportation. Fig. 1 highlights the sustainable parameters and STIs considered in the past work. Indicators are useful for comparing, tracking performance, and making judgments [27]. STIs have been proposed in the past by various studies, so Systematic Literature Review (SLR) is used to identify sustainable indicators for freight transportation [28]. The existing approach adopts a SLR to develop a comprehensive conceptual model using SEM and synthesize an emerging area of SFT, providing data to interpret prior findings for additional research on the subject [29,30]. The analysis of the literature revealed that prior studies on STI prioritized the ecological element of freight transportation, with only more recent studies demonstrating a more equitable distribution of triple bottom line indices [31,32]. Using SLR approach, it was discovered that majority of the studies on the sustainability of freight transportation listed following as the main impacts linked with freight transportation for triple bottom line of sustainability. In the former one economy, transportation costs, profitability, and government subsidies are considered. In the social aspect the prospects for wellbeing, equality, and justice are considered. While in the environmental aspect — noise pollution, resource utilization, and carbon emissions are the main parameters that have been used.

The risks involving freight transportation are well-described in literature, as shown by Fahimnia et al. [33] and Brandenburg and Rebs [34] via their independent research studies, where they could find 8 studies on sustainability risks. As a result, sustainability risks are an emerging topic and can be seen as a research gap for the present study [33]. Therefore, reducing the risks associated with sustainability is essential for the effective implementation of freight transportation [35]. More recently, it is recognized that the positive environmental impact of multimodality can be accelerated via the adoption of Information and Communication Technologies (ICTs) [36,37]. Gandhi et al. [38] adopted a triple bottom line analysis for sustainable performance evaluation considering the application of Indian Railways. A study by Davydenko et al. [39] suggested an energy efficient management for a smart sustainable freight transport. A framework to reflect the challenges to sustainable freight transportation was provided [40]. A case of Indian freight industry was considered to validate the study. The studies related to sustainable indicators and freight transportation are listed in Table 1:

To evaluate and assess the sustainability of the country's freight transportation systems, and the weight each indicator carries, STIs were identified and studied. In this study, risk indicators related to the viability of freight movement are included. In particular, the indicators used in this research belonged to one of the following 4 categories: economic, social, environmental, and risk. For these categories, 21 different criteria were identified bellowing to these four categories. In order to choose the criteria, it has been taken care of that criteria should be relevant, legitimate, autonomous and understandable. Table 2 provides the description of all these four features.

In this study, a balanced approach was undertaken to identify STIs in the following four categories: economic, social, environmental, and risk performance indicators. A triple bottom line concept was used to select these sixteen indicators and five sustainability parameters to analyze the sustainability of freight transportation across the nation. The respective categories and their indicators are discussed in detail below:-

2.1. Economic performance indicators

The majority of freight transportation costs are spent on the purchase of vehicles and the cost of fuel for the vehicles. Hence, the indicators such as "Average Price of the Vehicle" and the "Fuel Price" are used to measure the economic impact on freight transportation [3,43]. Tax breaks, tenders, and exchange programs implemented by the Government have an impact on how transportation companies make decisions. As a result, for the aforementioned criteria, the indicator such as "Government Subsidies" is chosen to determine the impact of these incentives on transportation companies and their decisions. To measure the viability and effectiveness of transportation operations, the indicator "Load Factor" is used [10].

2.2. Social performance indicators

Social value has become just as important to transport operators as economic value. Along with the financial standpoint, a company's overall working philosophy must take into account giving back to the community. A conceptual framework called the triple bottom line mandates that businesses evaluate both their financial and sustainability performance rather than concentrating only on their bottom line [44]. Hence, the indicator "Social Value" is used to measure an organization's social performance. Aside from adding social value, an organization must also fully feel responsible for its employees. The indicators such as "Safety", and "Employment Benefits" are used to assess the organization's commitment to employee well-being. Similarly, paying taxes and other Government levies for the benefit of society and an important aspect of an organization's social performance [45]. Hence, "Taxes" is used as an indicator.

2.3. Environmental performance indicators

Freight transportation accounts for 40% of air pollution in European cities [46]. Freight transportation contributes to the emissions of gases and particulate matter which are harmful to the environment and disturb the ecological balance of the planet. Trucks and trains responsible for freight transportation mainly run on diesel, contributing to NO_X and PM emissions [47,48]. Therefore, the indicator

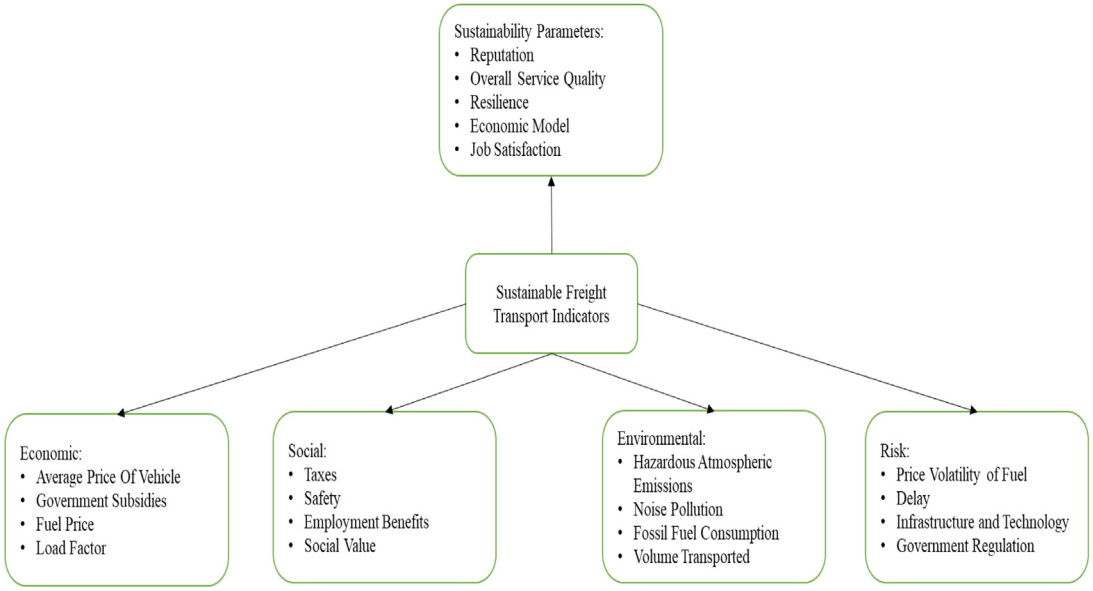

Fig. 1. Systematic diagram of Sustainable Transport Indicators (STIs).

Table 1
Studies related to sustainable indicators and freight transportation.

Author(s)	Review area	Considered indicator
Reefke and Sundaram [41]	Sustainability, Supply chain management	Risks related to freight transport has been considered.
Brandenburg and Rebs [34]	Sustainability, Supply chain management	Risks involved in transporting freight via cities is considered. Only economic considerations are made in this study.
Bandeira et al. [10]	Sustainable urban freight transportation in the city of Rio De Janeiro	Economic, social, and environmental factors are taken into account.
Fahimnia et al. [33]	Risk, Supply chain management	Highlights the importance of burgeoning field of sustainability risk analysis.
Brandenburg et al. [21]	Sustainability, Supply chain management	Social and Economic aspects are considered. However, Environmental indicators were majorly neglected in the paper.
Haghshenas and Vaziri [2]	Sustainable transport indicators for comparison	The study deals with freight transportation in the megalopolis.
Aven [42]	Risk management	The paper explores the link among risk and sustainability which needs to be further analysis.
Seuring and Mu [17]	Sustainability, Supply chain management	The accompanying sustainability concerns need to be taken into account in supplier evaluation plans.
Garg and Kashav [7]	Sustainability of freight Railways in India	Risks related to freight railway transportation in India.

Table 2
Identified sustainable freight transportation indicators and their explanation.

S. No.	Feature	Description
1.	Relevance	The indicator needs to represent a specific aspect of sustainable transportation.
2.	Legitimacy	Indicators must fully describe the issue there were intended to measure.
3.	Autonomous	The indicators should be autonomous of each other.
4.	Understandable	The signs should be simple to comprehend.

"Hazardous Atmospheric Emissions (CO, NO_X, and PM)" is used to measure the impact of freight transportation on emissions of these gases. Already, there is a significant association among both fossil energy consumption and greenhouse gas (GHG) emissions, we chose to assess the impact on natural resource usage and greenhouse emissions using the indicator "Fossil Fuel Consumption". Freight transport is also responsible for noise pollution in their vicinity, which causes discomfort and interference in people's and animals' everyday tasks; thereby, "Noise Pollution" is chosen as an indicator. The indicator "Volume Transported" is used to assess the environmental impact of freight transportation. A system that can move a larger volume in a smaller space consumes less space and has a lower overall impact on the environment.

2.4. Risk performance indicators

Sustainability risks are the biggest barrier in freight transportation [14]. They are inherent risks that can have direct consequences on the sustainability of the freight transportation system and can have negative effects on the system, therefore, disrupting the overall economic efficiency of the business model, and can have an impact on the implementation of the triple bottom line framework [15,49,50]. Government regulation does have an impact on how businesses run across the nation, thus by utilizing the indicator "Government Regulation", it could be assessed how Government regulations and bureaucracy affect transportation activities. Transportation Delays can cause an additional financial burden on the organization and can lead to further disruptions

in deliveries. Hence, the indicator "Delay" is used to evaluate the impact of transportation delays on sustainability. The development of a Freight Transportation System (FTS) would be difficult in the absence of adequate infrastructure and technological advancements. Therefore, the indicator "Infrastructure and Technology" is used. Unpredictable hikes in fuel prices have an impact on transportation costs and can severely disrupt the budget for freight transportation, hence the indicator "Price Volatility of Fuel" is used in this study.

The five sustainability parameters such as "Reputation", "Overall Service Quality", "Resilience", "Economic Model", and "Job Satisfaction" are used to measure the impact of STIs on sustainability. Table 3 represents the SFT indicators chosen in this study along with their descriptions and references.

Therefore, after performing an extensive literature study, a total 16 variables and 5 sustainability standards were established to analyze their impact on freight movement in the country. The risk, economic, social, and environmental indicators are consistent with literature analysis, whereas the sustainability metrics were built entirely from the ground up to assess the effects of the 16 indicators on the sustainability of the goods transportation, particularly on our 5 sustainability parameters.

3. Methodology

In this study, a SEM method is adopted to evaluate the data obtained from the survey. SEM is used to validate the conceptual model which has been theorized while identifying the key aspects for the sustainability of the nation's freight transportation system [61]. SEM is a robust statistical tool that integrates components of multiple regressions and factor analysis to evaluate both the structural and measurement attributes of conceptual models [62]. The level of interactions among the independent and dependent variables is specifically measured by SEM, demonstrating the impact of independent variables on the dependent variables. Hence, SEM is used in a wide range of social science applications [63]. SEM has been utilized in numerous studies such as [64], [65]. The steps of the methodology proposed in this study are detailed in the flowchart below [61] Fig. 2:

To examine their effects on national freight movement, 16 variables and 5 sustainability requirements were created for our model. In contrast to the 16 SFTIs that were developed and amended from the literature analysis; 5 new sustainability standards were added to evaluate their effects on the nation's freight transportation. Furthermore, the creation of SEM and analyzing its Model Fit Summary and the Regression Weights of the indicators to determine their impact on 5 standards to sustainability are the novelty in this study. This helps in determining the parameters importance in freight transportation in the country and enable both Government and private enterprises to adjust their strategies accordingly.

3.1. Data collection and questionnaire development

For this study, a survey form was shared with a variety of peer groups to gather their consenting responses to the questionnaire. The first step was to construct a survey form for the purpose of collecting the responses. There were 4 (economic, social environmental, and risk) categories and 21 questions on the survey form. The questionnaire is present in Appendix A. The questions were then answered adopting a five point likert scale ranging from equal importance to extreme importance to determine the recipient's perception of the sustainability of freight transportation. The 5 point likert scale of relative importance is mentioned below in Table 4.

3.1.1. Data validation

The data validation process comprises of three stages:

3.1.1.1 Data preparation: The first stage in the data validation process was to review the variables in the data file and determine their correct labels, measurement levels, and values. After collecting the data using an SPSS Statistics data sheet, the data must then be cleaned to eliminate duplicate and inaccurate responses. Invalid responses, such as those referring to errors, were given a null value to reduce their influence on the result while preserving the integrity of the data collection.

3.1.1.2 Validation: Invalid values were identified after fundamental checks and comparisons with established validation criteria were performed. The data was automatically rectified after it was determined to be erroneous. The 5-point Likert scale, displayed in Table 4 was used to define the range of values. According to the SPSS validation criterion, any values that did not fall inside the aforementioned scale and any missing values were automatically ruled invalid and assigned null values or in certain cases were manually assigned to ensure a good model fit.

3.1.1.3 Model preparation: The cleansed data was utilized to create the SEM and then for additional analysis after data cleaning and validation.

3.1.2. Sample size

The required sample size for SEM is not specified. According to that a priori Sample Size method was used to calculate the required sample size, the formulas given below in Eqs. (1) and (2) were used [66]:

$$F\ x;\ ;\ ^2 = \frac{1}{2}[1 + \mathrm{erf}(\frac{x -}{\tilde{u} -})] \quad (1)$$

where is the standard deviation, is the mean and the erf is the error function.

The error function is given by the formula in the Eq. (2) below:

$$\mathrm{erf}\,(x) = \frac{2}{\tilde{u}} \int e^{-t^2} : dt \quad (2)$$

Hence, the adoption of above two equations reflects our sample size to be 173 and rounding it to 174. To accommodate a greater sample size and a wide range of demographics, responses were collected from a variety of different audiences to get a good model fit and to include calculations for the entirety of the population. The sample size for the SEM consists of responses from the three groups comprising of academician, industrialist, and students. The responses were collected via. a survey form and the results were used to construct SEM. The respondents were required to fill out a survey form for each of the 21 Sustainable Transportation Indicators (STIs), and their responses were then gathered. An average of 8.26 cases/observations per indicator was obtained. The responses originate from a range of data sources, with student responses serving as the foundation of our statistics. To ensure that the model accurately represents the entire country and that the survey's finding, the questionnaire was sent to respondents from a wide range of demographics and industries across the nation.

3.2. Structural equation model approach

In this study, SEM methodology is used for validation of the hypothesized conceptual model. SEM is a method for estimating a series of dependence relationships between a comprehensive framework and structures defined by numerous measured variables and integrated into a single model [61]. The definition of a theoretical model based on validated research is the first step in the SEM analysis. The method for measuring the structures under examination is specified by the researcher, who then gathers the required input data and conducts the SEM analysis. The technique consists of evaluations of the constructs' validity, reliability, and discriminant power, as well as estimations of the model's parameters and, lastly, statistics relating to how well the model fits the data overall (do cite: [67]). The path diagram of SEM is presented in Fig. 3:

Table 3
The sustainable freight transportation indicators.

Category	Indicators	Description	References
Economic performance indicators	Price of the Vehicle	The transportation of freight relies on vehicles. The primary expense for transportation firms is the cost of the vehicle. The cost of ownership directly impacts the company's business strategy. Newer, more efficient vehicles cost more money.	Lindholm [46], Mangla et al. [50] and Stephenson et al. [5]
	Government Subsidies	The government implemented tax advantages, competitions, and exchange programs to help the industry.	Haghshenas and Vaziri [2]
	Fuel Price	The cost of transportation is influenced by fuel prices. Higher the fuel price lowers would be the profits.	Indicators et al. [51] and Awasthi and Chauhan [52]
	Load Factor	The load factor also called occupancy rate is an economic indicator that is used to measure the effectiveness of transportation operations and is consequently linked to the operation's viability.	Ii [3] and Bandeira et al. [10]
Social performance indicators	Taxes	Government taxes, such as tolls, corporate taxes, and GST, generate revenue for the government, which can be used to improve infrastructure and healthcare services, as well as to improve our society. Thus, paying proper taxes is critical for us for the government to provide us with such benefits.	Haghshenas and Vaziri [2] and Shankar et al. [14]
	Safety	One of the most crucial factors that businesses should consider in the well-being of their employees is employee safety. Companies have an obligation to their workforce to ensure their employees' safety.	Haghshenas and Vaziri [2]
	Employment Benefits	It is important to offer employee perks including a consistent wage, decent housing, and paid time off for illness.	Study et al. [53] and Kousadikar and Singh [54]
	Social Value	Social value is as important to companies as an economic benefit. Along with economic aspects, a firm's working philosophy must include giving back to the community. Accountability, fairness, and humanity are examples of social values that improve the public reputation and send a positive personal message.	Awasthi and Chauhan [52], Study et al. [53] and Stank and Goldsby [55]
Environmental performance indicators	Hazardous Atmospheric Emissions	The majority of air pollution is caused by vehicles. Diesel trains and buses are responsible for the emission of NO_X and PM, which are harmful and increase the risk of cancer.	Fridell et al. [56], Wi [57] and Taylor et al. [58]
	Noise Pollution	Freight transport also contributes to noise pollution. Noise pollution can cause deafness, depression, anxiety, and a variety of other health problems. As a result, freight companies must use vehicles that emit less noise pollution.	Fridell et al. [56], Wi [57] and Taylor et al. [58]
	Fossil Fuel Consumption	The use of fossil fuels has a direct effect on CO, NO, and PM emissions. These raise the carbon footprint and accelerate the process of global warming. They are detrimental to health and have been related to various malignancies.	Haghshenas and Vaziri [2] and Bandeira et al. [10]
	Volume Transported	The "Volume Transported" indicator shows how the operation impacts the immediate landscape. A system that can transport additional mass (or volume) in a limited footprint occupies less space and typically has a less overall impact. Together with the loading capacity in weight, the load capacity in volume is often used as a performance indicator (tons).	Bandeira et al. [10]
Risk performance indicators	Price Volatility of Fuel	Unpredictable hikes in fuel prices have an impact on transportation costs and can severely disrupt the budget for freight transportation. The higher the fuel price, the fewer profits the companies are earning.	Bandeira et al. [10]
	Government Regulation	Regulations made by the government serve as useful guidelines that set the limits of acceptable conduct. Government regulations have a major impact on how businesses function.	Gan [59], Stephenson et al. [5] and Abbasi and Nilsson [18]
	Infrastructure and Technology	The development of a freight transportation system would be difficult in the absence of adequate infrastructure and technological advancements, and the use of antiquated vehicles with obsolete technology makes the adoption of sustainability even more challenging.	Shankar et al. [14]
	Delay	Risk due to the detrimental effects of transportation delays, such as delays in the delivery of goods, and requiring additional visits for deliveries affects the viability of the freight transfer system.	Shankar et al. [14]
Sustainability parameters	Reputation	A positive reputation is crucial for attracting clients. Better employees are drawn to companies with strong good reputations. Given that they are regarded as providing superior value, they can frequently charge more. They have more loyal customers who buy a larger range of goods and services.	Haghshenas and Vaziri [2], Suki and Suki [60]
	Overall Service Quality	An organization's overall service quality is determined by evaluating how well it meets customer expectations in terms of the services it provides. Hence, good service quality is important for drawing in more customers.	Bandeira et al. [10]
	Resilience	An organization's resilience is its capacity to anticipate, prepare for, respond to, and adapt to changes occurring and unexpected interruptions to flourish.	Shankar et al. [14]
	Economic Model	A successful economic model is crucial to the long-term viability of an organization. It helps the business to operate profitably going forward while also enabling them to work effectively and efficiently within sustainable economic, social, and environmental goals.	Shankar et al. [14]
	Job Satisfaction	An indicator of staff motivation with their employment is job satisfaction. Higher job satisfaction for employees requires a suitable working environment and better employment perks. A positive work atmosphere attracts better employees and raises performance.	Haghshenas and Vaziri [2]

Fig. 2. The key steps of the presented research study.

Table 4
The 5 point likert scale of relative importance.

Influence score	Linguistic variable	Explanation
1	Equal Importance (EI)	Two criteria contribute equally to the object
3	Moderate Importance (MI)	Judgment favors one criterion over another
5	Strong Importance (SI)	Judgment strongly favors one criterion over another
7	Very Strong Importance (VSI)	A criterion is favored very strongly over another
9	Extreme Importance (EXI)	The evidence favoring one criterion over another

The Structural Regression Model (SRM), which illustrates interactions among latent variables, and the Measurement Model, which defines and validates latent variables as linear functions of observable variables, are two sub-models that together make up the phrase SEM. A set of simultaneous regression equations connects the two sub-models as highlighted in the study [68]. Finally, the model is tested and the model fit goodness is determined to analyze how the model fits the sampled data [69]. Later, the model is checked in several categories such as the Goodness of Fit Index (GFI), the Root Mean Square Residual (RMR), the Adjusted Goodness of Fit Index (AGFI) and the Root Mean Square Error of Approximation (RMSEA).

4. Results and discussion

The results of the existing study highlights the significance of the STIs in the feasibility of freight transportation in the country and how organizations can benefit by aligning their goals and strategy for resource minimization and profit maximization. A company can boost revenue and profit by lowering operating costs. By continuing to use the sustainable practices outlined above, they can also raise their profit margins. This study analysis the impact of the mentioned transportation indicators and assess their impact on sustainability of freight transport. The study contributes towards adoption of sustainability in the domain of freight transportation for better sustainability, economic growth and lower the fuel and emission's cost along with reducing the carbon footprint. The findings from the study can help the organization move toward a sustainable model that is more in line with the United Nation (UN) Sustainable Development Goals (SDGs).

In this study, the relationships between the various variables are examined, and their impact on transportation sustainability is calculated using SPSS AMOS. The aforementioned path model was created on AMOS, and its results were examined. The list of variables used in the above model and their labels are presented in Appendix B.

In this model, a dependent interaction is reflected among all the endogenous construct (Sustainability Parameters) and the exogenous construct (Economic, Social, Environmental and Risk Indicators). Furthermore, to improve the model fit a correlational relationship is shown among all the exogenous constructs in Fig. 4.

The model fit summary assesses the variance among the observed and model-derived data by using correlation and covariance matrices in Table 5. The parameters observed during model fit are illustrated in Appendix C.

- HI 90: - Higher boundary of a 90% confidence interval of the RMSEA.
- PCLOSE: - P-value of the null hypothesis.